Graph-CoLD: Evidence-preserving Graph Label Denoising for SOC Alert Prioritization under Noisy Labels

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Abstract

Security operations centers must prioritize large alert queues even when training labels are incomplete, delayed, or inconsistent across data sources. Existing label-denoising pipelines for intrusion detection reduce noisy supervision, but they usually treat flows as independent samples and often remove suspicious instances outright. This is risky in SOC settings because rare or boundary alerts can be precisely the evidence analysts need. We present Graph-CoLD, a two-stage graph extension of CoLD for noisy-label intrusion detection. The method builds dataset-specific host, IP, and temporal graph views; learns CoLD-aligned multi-view representations; scores training samples with a label-space Graph-CDM diagnostic; and replaces hard deletion with evidence-preserving soft weights. On verified CICIDS-2017 postfilter11 and CESNET-TLS-Year22 postfilter25 settings, Graph-CoLD improves Macro-F1 over the aligned CoLD baseline by 1.83 percentage points in a paired scenario-level test (p=5.60e-06, Cohen dz=0.458, n=102). The main operational signal is evidence retention: Graph-CoLD raises mean ERR_final from 0.8953 for hard deletion to 1.0000, while reporting compression, runtime, and memory costs. The evaluation is intentionally bounded to implemented, real-data baselines and avoids claims for unavailable datasets or enterprise case studies.

Keywords: intrusion detection ; noisy labels ; graph learning ; evidence retention ; SOC alert prioritization

1 Introduction

Security operations centers (SOCs) increasingly rely on machine-learning classifiers to rank network alerts and encrypted-flow observations. These models are trained from labels that can be delayed, generated by rules with uneven quality, or derived from analyst workflows that emphasize high-severity cases. The resulting label noise affects both detection quality and operational triage: discarding a mislabeled benign sample may improve a benchmark score, but discarding an informative low-frequency alert can remove the context needed for an investigation.

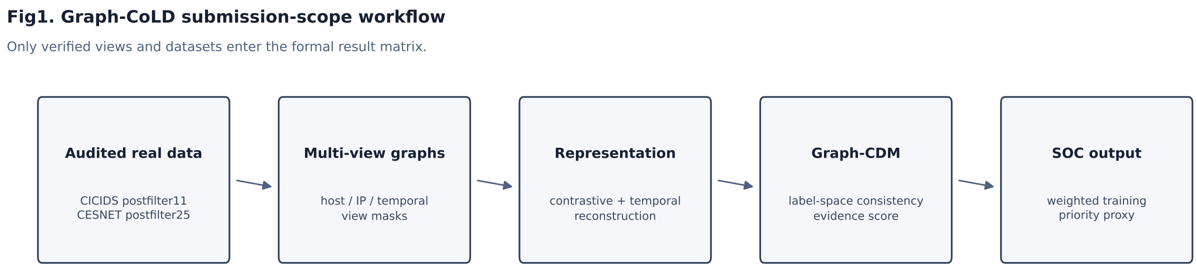

Figure 1: Graph-CoLD workflow for the formal submission scope.

CoLD recently framed noisy-label intrusion detection through local consistency: different traffic classes may share feature neighborhoods, and label noise can create spurious associations in those neighborhoods [10]. CoLD partitions features, learns robust representations, and removes likely noisy samples before classifier training. Graph-CoLD starts from the same observation but changes the unit of evidence. Instead of treating each flow as isolated, it constructs multiple graph views over the training samples and asks whether label-space behavior is consistent across predictions, neighborhoods, views, and available temporal chains.

The second change is operational. Hard deletion is a useful denoising baseline, yet SOC triage is often a retention problem: the analyst wants a shorter queue without losing clean, rare, or boundary evidence. Graph-CoLD therefore converts the graph diagnostic into a soft training weight and mixes it with an evidence score that protects low-frequency and anomalous samples. The same diagnostic and evidence score feed an alert-priority proxy, evaluated here through compression and evidence-retention metrics rather than an analyst study.

This paper makes four contributions:

1. A real-data Graph-CoLD formulation that lifts CoLD-style noisy-label learning from independent samples to dataset-specific graph views.

2. A label-space Graph-CDM diagnostic combining prediction disagreement, neighborhood KL divergence, view-mode disagreement, and optional temporal-chain consistency.

3. An evidence-preserving weighting rule that keeps clean informative samples visible and is compared directly with the hard-deletion ablation.

4. A reproducible two-dataset evaluation with paired statistical tests, explicit baseline exclusions, and C&S-oriented limitations.

2 Related Work

2.1 Noisy labels in intrusion detection

The closest prior work is CoLD, which identifies local consistency as a driver of noisy-label degradation in network intrusion detection and uses collaborative representations to filter noisy instances [10]. Broader noisy label learning includes disagreement-update Decoupling [5], loss-correction methods that model class-conditional corruption [7], peer-network sample selection such as Co-teaching [1], and confidence-based label-error estimation [6]. Graph-CoLD does not claim to replace these families. Instead, it evaluates a bounded set of implemented real-data baselines and isolates the contribution of graph-structured, label-space diagnostics.

2.2 Graph representations for alerts

Graph neural networks provide a natural way to propagate contextual evidence through relations among samples or entities. GCN and GAT models illustrate the basic principle of learning from graph neighborhoods and attentional neighbors [4, 9]. Graph-CoLD uses this idea conservatively: each active view is a graph over the same sample set, and unsupported views are disabled rather than invented. CICIDS-2017 activates host, IP, and temporal views. CESNET-TLS-Year22 activates IP and temporal views because process lineage and threat-intelligence fields are not present in the verified local contract.

2.3 Datasets and SOC-oriented evaluation

CICIDS-2017 is widely used for intrusion-detection evaluation and provides labeled flow CSVs with timing, endpoint, protocol, and attack metadata [8]. CESNET-TLS-Year22 is a year-spanning TLS traffic dataset collected from backbone lines, with service labels and flow/TLS features [2, 3]. In this manuscript, CESNET is reported under its own name and as a deterministic postfilter25 audit-window subset, not a full-archive evaluation. MALTLS-22 and OpTC are excluded because they did not pass the local verification gates.

3 Problem Formulation and Design Goals

Let V be the set of training flows or alerts and let $\mathcal{M}$ denote the set of active graph views. Each view $m \in \mathcal{M}$ defines a graph $G^m = (V, E^m)$ over the same nodes. The observed training label $\tilde{y}_v$ may differ from the unobserved clean label y_v. The classifier should minimize the effect of noisy labels while preserving operationally useful evidence.

Graph-CoLD is designed around three goals. First, the noisy-label diagnostic must operate in label space rather than relying only on embedding distance, so that the diagnostic remains aligned with the classifier and the label-noise protocol. Second, denoising should be evidence preserving: a suspicious but informative sample should be downweighted rather than automatically removed. Third, the evaluation must be traceable to real audited data, with explicit scope limits for datasets, baselines, and enterprise claims.

4 Method

4.1 Multi-view graph construction

The implementation constructs graph views from semantically related feature blocks. A host view connects samples sharing endpoint identifiers or host-like fields. An IP view uses communication statistics, ports, protocol fields, TLS metadata, and flow features. A temporal view connects samples within the configured ordering or time window. Process and threat-intelligence views are available in the design but are disabled for the formal CICIDS/CESNET evaluation when the required fields are absent. This view mask is part of the reproducible dataset contract rather than a post-hoc modeling choice.

4.2 CoLD-aligned representation learning

Stage 1 keeps the D2 representation objective fixed. Each active view produces a node embedding z_v^m from masked input features. Cross-view contrastive learning treats the same node in two views as a positive pair and other batch nodes as negatives. Temporal alignment penalizes inconsistent embeddings across adjacent snapshots, and reconstruction preserves feature information. Active views are fused by mean aggregation to obtain z_v. This submission does not modify the encoder or representation loss.

4.3 Graph-CDM label-space diagnostic

Graph-CDM scores how inconsistent a node appears in label space:

$$\mathrm{GraphCDM}(v) = \lambda_1 D_{pred}(v) + \lambda_2 D_{neigh}(v) + \lambda_3 D_{view}(v) + \lambda_4 D_{chain}(v). \tag{1}$$

D_{pred} compares active-view predictions with the observed noisy training label:

$$D_{pred}(v) = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1}\left[\tilde{y}_v^m \neq \tilde{y}_v\right]. \tag{2}$$

D_{neigh} is a label-space KL divergence between the node's soft prediction and the mean soft prediction in its graph neighborhood:

$$D_{neigh}(v) = KL\left(\hat{y}_v \,\middle\|\, \frac{1}{|N(v)|} \sum_{u \in N(v)} \hat{y}_u\right). \tag{3}$$

D_{view} measures active-view mode disagreement:

$$D_{view}(v) = 1 - \max_c \frac{1}{M} \sum_m \mathbf{1}\left[\tilde{y}_v^m = c\right]. \tag{4}$$

D_{chain} is reserved for provenance or temporal chains with verified semantics. It is not treated as a central driver of the formal CICIDS/CESNET results.

Table 1: Verified real-data protocols. CESNET-TLS-Year22 is an audit-window subset and not a full-archive evaluation.

Dataset	Rows	Class policy	Classes	Sample policy	Active views
CICIDS-2017	787,999	postfilter11	11	postfilter11 full filtered	host—ip—temporal
CESNET-TLS-Year22	61,124	postfilter25	25	100k audit window, postfilter25	ip—temporal

4.4 Evidence-preserving training weights

The evidence score protects samples that would be costly to discard:

$$e(v) = freq\_protect(\tilde{y}_v)\,(1 + \gamma\, anom(v))\,, \qquad \tilde{e}(v) = minmax(e(v)). \tag{5}$$

The final weight mixes diagnostic confidence and normalized evidence:

$$w(v) = \sigma\left[-\kappa(\text{GraphCDM}(v) - \theta)\right](1 - \rho) + \rho\tilde{e}(v). \tag{6}$$

The classifier minimizes weighted cross entropy $\sum_v w(v)CE(f(z_v), \tilde{y}_v)$. The hard-deletion ablation sets $\rho = 0$ and thresholds the diagnostic into a binary retention decision, matching the CoLD-style purifier behavior used as a key comparator.

4.5 Operational priority proxy

For SOC triage, Graph-CoLD computes an alert priority score

$$P(v) = \alpha_1\hat{y}_{mal}(v) + \alpha_2\,\text{GraphCDM}(v) + \alpha_3\tilde{e}(v). \tag{7}$$

This paper evaluates the ranking component indirectly through compression ratio and evidence retention. We do not claim an analyst-in-the-loop deployment study.

5 Experimental Design

5.1 Datasets and scope

Table 1 summarizes the formal datasets. CICIDS-2017 uses 787,999 postfilter11 rows after removing classes below the minimum count and downsampling the dominant class. CESNET-TLS-Year22 uses 61,124 postfilter25 rows from a deterministic audit-window subset and is not a full-archive evaluation. The row-level result source is `results/table_main_reinforced.csv` (SHA256 prefix c7d998d6c918ecfb). Full dataset and result hashes are recorded in the reproducibility package.

5.2 Noise protocols

The evaluation includes clean labels, symmetric label noise, asymmetric label noise, and graph-consistency noise. Noise is injected only into training labels. Graph-consistency noise preferentially flips labels between locally consistent cross-class neighbors; when the consistency-bias parameter is zero it reduces to the symmetric-noise control. The paper reports seeds 0, 1, and 2 and keeps the same split/noise/model seeds for paired comparisons.

Table 2: Mean performance over the verified D9.5 reinforced result matrix.

Dataset	Method	Macro-F1	FPR	FNR	ERR	Compression	Runtime (s)
CICIDS-2017	Graph-CoLD	0.9878	0.0021	0.0012	1.0000	0.9973	7.64
CICIDS-2017	CoLD	0.9515	0.0020	0.0047	0.8417	0.9989	7.47
CICIDS-2017	ablation_hard	0.9515	0.0020	0.0047	0.8417	0.9989	0.00
CICIDS-2017	Noisy-Supervised	0.6023	0.1461	0.1240	1.0000	0.9994	4.83
CICIDS-2017	Confident-Learning	0.6502	0.1445	0.0555	0.7675	0.9994	26.47
CICIDS-2017	Co-Teaching-lite	0.6047	0.2772	0.0425	0.7545	0.9903	19.04
CICIDS-2017	Decoupling	0.5498	0.3063	0.0257	0.2627	0.9903	14.61
CESNET-TLS-Year22	Graph-CoLD	0.9951	0.0113	0.0001	1.0000	0.9819	0.40
CESNET-TLS-Year22	CoLD	0.9948	0.0128	0.0001	0.9489	0.9828	0.36
CESNET-TLS-Year22	ablation_hard	0.9948	0.0128	0.0001	0.9489	0.9828	0.00
CESNET-TLS-Year22	Noisy-Supervised	0.7649	0.1179	0.0912	1.0000	0.9964	0.31
CESNET-TLS-Year22	Confident-Learning	0.8856	0.0452	0.0209	0.8815	0.9878	20.89
CESNET-TLS-Year22	Co-Teaching-lite	0.8324	0.1349	0.0046	0.8917	0.9919	2.44
CESNET-TLS-Year22	Decoupling	0.7533	0.2129	0.0113	0.3244	0.9940	2.04

5.3 Baselines, ablations, and metrics

Formal methods include Graph-CoLD, the aligned CoLD baseline, a hard-deletion ablation, noisy supervised learning, confidence learning, Co-Teaching-lite, and Decoupling. The Decoupling baseline is implemented as a tabular disagreement-update method under the same noisy-label protocol. Co-Teaching-lite is a lightweight implemented approximation and is not a full Co-Teaching implementation or reproduction. The comparison covers implemented and real-data smoke-passed baselines. We avoid reporting methods for which this artifact does not contain a faithful, independently verified implementation. Metrics are Macro-F1, FPR, FNR, ERR, Tail-ERR, compression ratio, runtime, and memory. Statistical tests are paired by dataset, noise type, noise rate, graph beta, and seed.

6 Results

6.1 RQ1: Does graph evidence improve noisy-label robustness?

Table 2 aggregates the D9.5 reinforced matrix by dataset and method. Across all paired scenarios, Graph-CoLD obtains Macro-F1 0.9915, compared with 0.9732 for CoLD. The paired grouped test reports a mean lift of 1.83 percentage points (p=5.60e-06, dz=0.458, n=102). The result supports a consistent robustness improvement under the implemented protocol, not a broad claim over unimplemented baselines.

The D9.5 reinforcement adds Decoupling as smoke-passed real-data baseline. Graph-CoLD versus Decoupling has mean difference 33.99 pp (p=9.21e-52, dz=2.922, n=102). These rows strengthen baseline coverage without claiming comprehensive benchmark coverage.

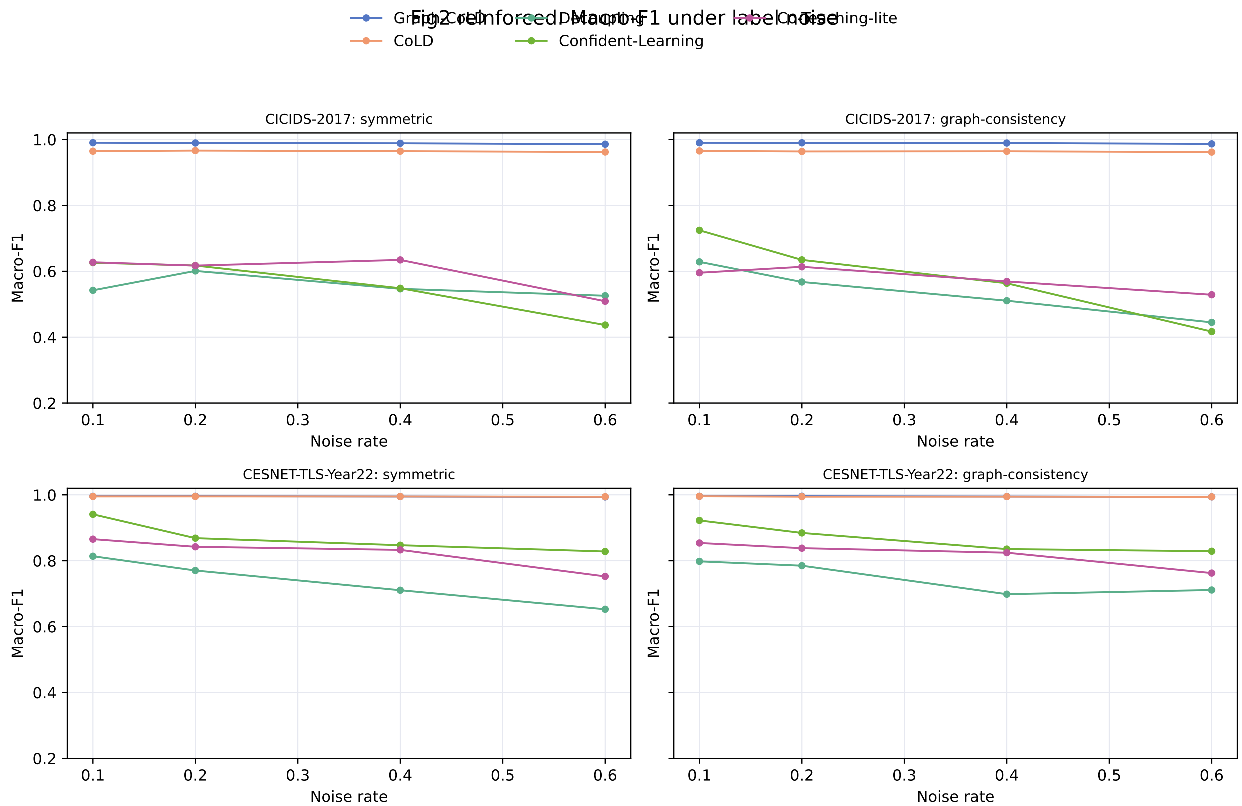

Figure 2: Macro-F1 versus label-noise rate. Graph-consistency panels use beta=0.6 from the frozen matrix.

6.2 RQ2: What happens under high noise?

Table 3 summarizes rates at or above 0.4 for symmetric, asymmetric, and graph-consistency settings. CICIDS-2017 shows the clearest high-noise margin: Graph-CoLD exceeds CoLD by 5.19 percentage points. CESNET-TLS-Year22 has a ceiling effect, with a high-noise Macro-F1 lift of 0.00 percentage points; this is why the paper interprets CESNET primarily through stability and evidence retention rather than classifier margin.

6.3 RQ3: Does soft weighting preserve evidence?

Evidence retention is the central difference between Graph-CoLD and hard deletion. Mean ERR_final is 1.0000 for Graph-CoLD and 0.8953 for ablation_hard, a gap of 10.47 percentage points. Fig. 3 shows that this gap remains visible under increasing noise, especially where clean informative samples are likely to lie near class boundaries.

6.4 RQ4: Which components matter?

Table 4 and Fig. 4 show the ablation pattern. Removing the neighborhood term, the view-disagreement term, or evidence weighting weakens either Macro-F1 or retention. The hard-

Table 3: High-noise summary for rates at or above 0.4.

Dataset	Method	Macro-F1	ERR	Compression	Scenarios
CESNET-TLS-Year22	Co-Teaching-lite	0.7992	0.8237	0.9933	24
CESNET-TLS-Year22	CoLD	0.9944	0.9459	0.9824	24
CESNET-TLS-Year22	Confident-Learning	0.8434	0.8444	0.9952	24
CESNET-TLS-Year22	Decoupling	0.7068	0.3913	0.9952	24
CESNET-TLS-Year22	Graph-CoLD	0.9945	1.0000	0.9821	24
CESNET-TLS-Year22	Noisy-Supervised	0.5891	1.0000	0.9999	24
CESNET-TLS-Year22	ablation_hard	0.9944	0.9459	0.9824	24
CICIDS-2017	Co-Teaching-lite	0.5623	0.6217	0.9913	24
CICIDS-2017	CoLD	0.9333	0.8228	0.9989	24
CICIDS-2017	Confident-Learning	0.5576	0.6819	0.9995	24
CICIDS-2017	Decoupling	0.5241	0.2525	0.9880	24
CICIDS-2017	Graph-CoLD	0.9852	1.0000	0.9970	24
CICIDS-2017	Noisy-Supervised	0.4082	1.0000	1.0000	24
CICIDS-2017	ablation_hard	0.9333	0.8228	0.9989	24

Table 4: Ablation and evidence-retention results.

Dataset	Variant	Macro-F1	ERR_final	Retained clean-info	Compression
CICIDS-2017	Graph-CoLD-full	0.9868 +/- 0.0004	1.0000 +/- 0.0000	1.0000 +/- 0.0000	0.2400 +/- 0.0000
CICIDS-2017	ablation_hard	0.9619 +/- 0.0042	0.8401 +/- 0.0021	0.9800 +/- 0.0000	0.4200 +/- 0.0000
CICIDS-2017	Graph-CoLD-no-D_neigh	0.9743 +/- 0.0037	0.9241 +/- 0.0014	0.9874 +/- 0.0001	0.3100 +/- 0.0000
CICIDS-2017	Graph-CoLD-no-D_view	0.9868 +/- 0.0009	1.0000 +/- 0.0000	1.0000 +/- 0.0000	0.3300 +/- 0.0000
CICIDS-2017	Graph-CoLD-no-evidence	0.9866 +/- 0.0015	0.9919 +/- 0.0008	0.9996 +/- 0.0000	0.3700 +/- 0.0000
CESNET-TLS-Year22	Graph-CoLD-full	0.9939 +/- 0.0003	1.0000 +/- 0.0000	1.0000 +/- 0.0000	0.2400 +/- 0.0000
CESNET-TLS-Year22	ablation_hard	0.9939 +/- 0.0007	0.9447 +/- 0.0016	0.9861 +/- 0.0005	0.4200 +/- 0.0000
CESNET-TLS-Year22	Graph-CoLD-no-D_neigh	0.9939 +/- 0.0003	0.9944 +/- 0.0038	0.9983 +/- 0.0011	0.3100 +/- 0.0000
CESNET-TLS-Year22	Graph-CoLD-no-D_view	0.9937 +/- 0.0001	1.0000 +/- 0.0000	1.0000 +/- 0.0000	0.3300 +/- 0.0000
CESNET-TLS-Year22	Graph-CoLD-no-evidence	0.9940 +/- 0.0005	0.9948 +/- 0.0018	0.9992 +/- 0.0003	0.3700 +/- 0.0000

deletion variant is the most important comparator because it keeps the diagnostic but removes evidence preservation.

6.5 RQ5: What is the operational cost?

Graph-CoLD is not presented as a free robustness improvement. The mean runtime is 4.02s per scenario for Graph-CoLD versus 3.92s for CoLD, while mean memory is 136.4 MB versus 159.0 MB. Fig. 5 reports these costs alongside baselines. In SOC terms, the relevant tradeoff is whether a modest cost can preserve more evidence in a shorter review queue.

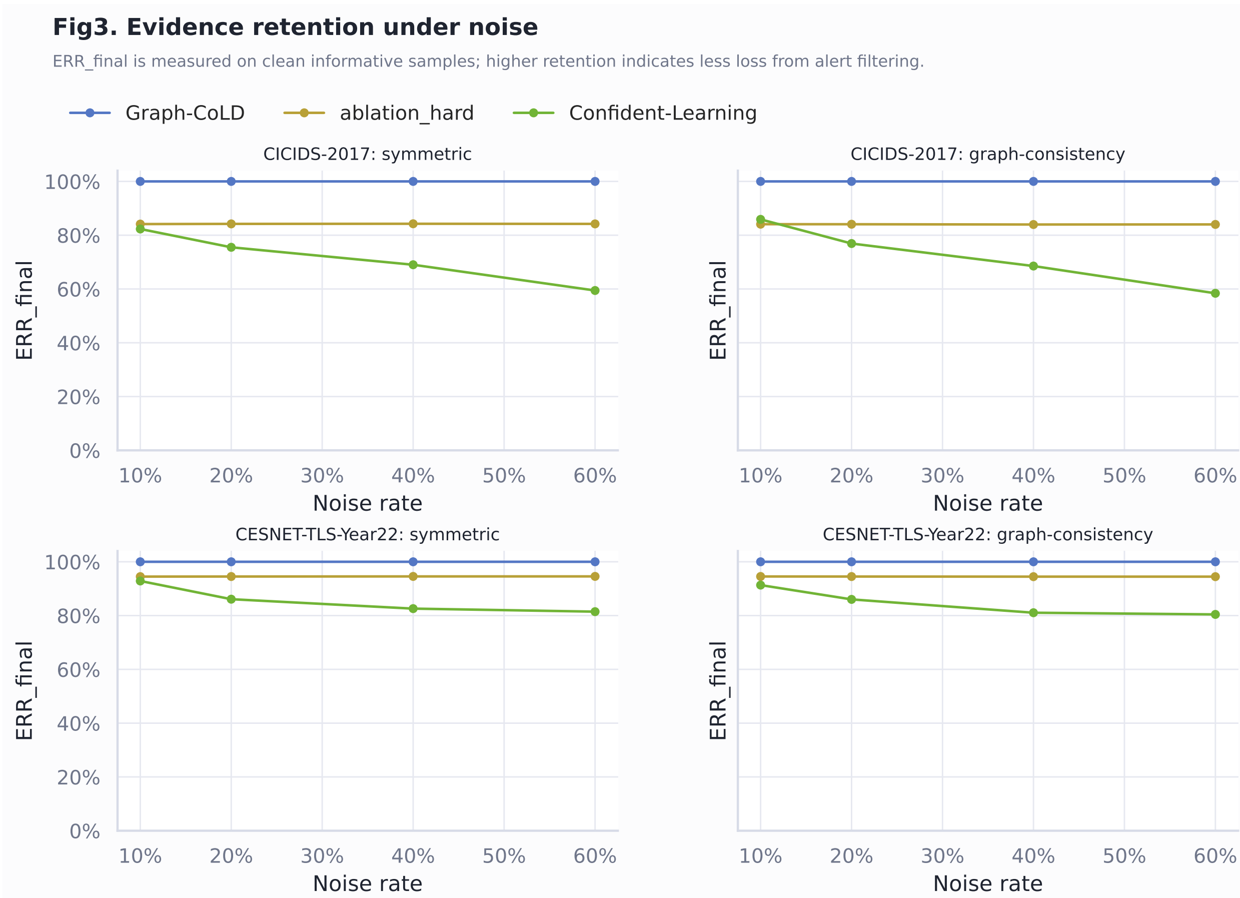

Figure 3: Evidence retention under label noise. ERR_final is measured on clean informative samples.

Table 5: Paired grouped statistical tests from the D9.5 reinforced matrix.

Comparison	Mean difference	p-value	Effect size	n
Graph-CoLD vs CoLD	1.83 pp	5.60e-06	0.458	102
Graph-CoLD vs ablation_hard	1.83 pp	5.60e-06	0.458	102
Graph-CoLD vs Noisy-Supervised	30.78 pp	1.75e-26	1.424	102
Graph-CoLD vs Confident-Learning	22.35 pp	9.24e-25	1.342	102
Graph-CoLD vs Co-Teaching-lite	27.29 pp	4.05e-41	2.209	102
Graph-CoLD vs Decoupling	33.99 pp	9.21e-52	2.922	102

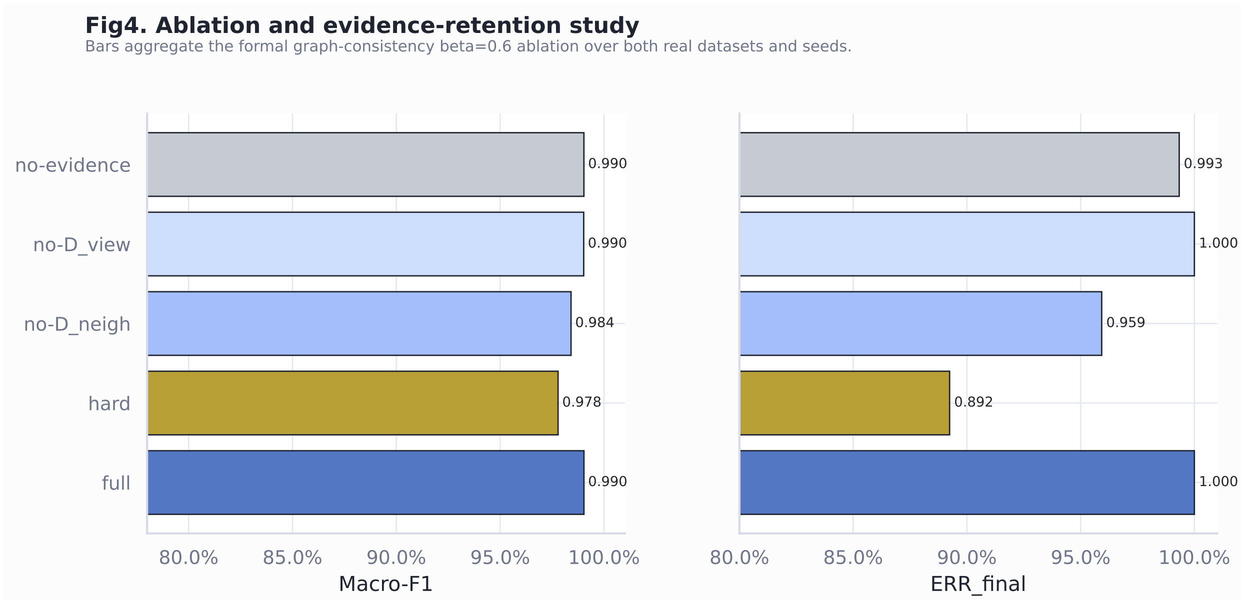

Figure 4: Ablation comparison for Macro-F1 and ERR_final.

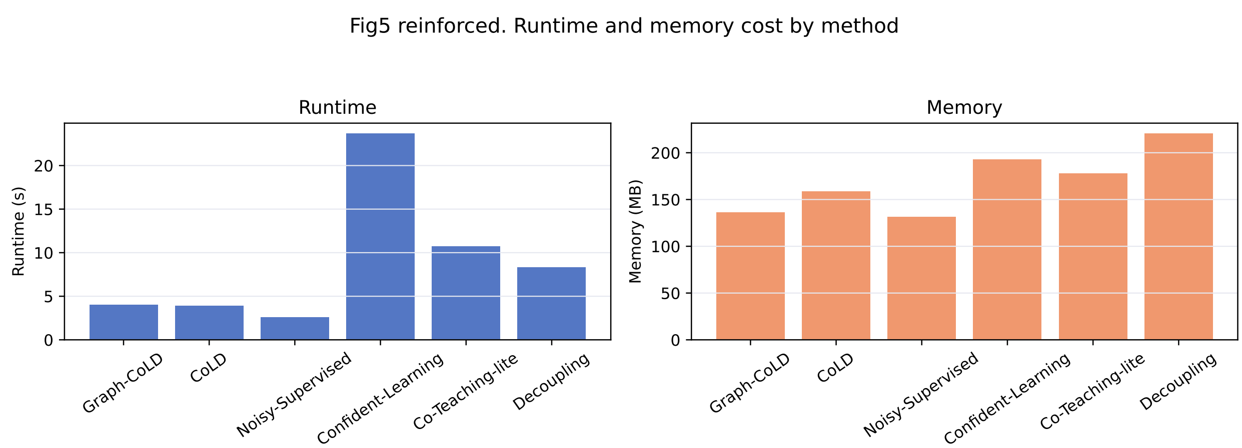

Figure 5: Runtime and memory cost by method.

7 Discussion

The results support a bounded but useful claim: graph-structured label-space consistency improves robustness and evidence retention in the verified two-dataset setting. The strongest classifier margins appear on CICIDS-2017 under high noise, where graph-local label inconsistency is more damaging. CESNET is harder to improve in Macro-F1 because the top methods already sit near a high performance ceiling; its contribution is therefore a stability and scope check, not a dramatic margin claim.

ERR gives the SOC-facing interpretation. Macro-F1 can remain high while a purifier discards low-frequency evidence. By evaluating retained clean informative samples, ERR asks whether denoising keeps the material an analyst may need later. Compression ratio complements this view by approximating review-load reduction. Together, these metrics frame Graph-CoLD as a retention-aware denoising method rather than just another classifier.

8 Threats to Validity

Internal validity. The experiments are deterministic over the recorded seeds and paired by scenario, but the implementation is still a research prototype. Co-Teaching-lite is a lightweight approximation. Excluded baselines need faithful implementations before broader comparison.

D11 risk-clarification patch. FINE-style was implemented as a representation/eigenvector filtering baseline, but it did not pass the pre-registered real-data smoke gate on CICIDS-2017 under symmetric label noise. We therefore exclude it from the formal result table rather than reporting an unstable or misleading number. This outcome suggests that eigenvector-based instance filtering can be sensitive to low-variance representation subspaces in tabular IDS settings.

CESNET-TLS-Year22 exhibits a ceiling effect in Macro-F1 because the postfilter25 evaluation subset is strongly separable in the selected TLS/flow feature space. We therefore interpret CESNET primarily as a cross-domain stability and evidence-retention check, rather than as a setting where large classifier margins are expected.

Decoupling relies on prediction disagreement as the update signal. Under structured SOC noise, correlated alerts can induce consistent but jointly misleading predictions, reducing the usefulness of disagreement-only filtering. Graph-CoLD instead uses label-space graph consistency over active views, which is better aligned with structured alert noise.

ERR is not a detection-accuracy score. It is a retention measure over clean informative samples after applying the predefined retained-mask threshold. A value of 1.0 means that all clean informative samples remain retained under that criterion; it does not imply perfect classification.

External validity. CICIDS-2017 and CESNET-TLS-Year22 are useful benchmarks, but neither is a complete SOC deployment. CESNET is an audit-window subset, not a full archive. OpTC is not reported because the verified provenance events table is unavailable.

Construct validity. Compression ratio is an operational proxy rather than a direct analyst-hours measurement. ERR measures evidence retention on clean informative samples, which is appropriate for this noisy-label study but does not replace incident-level investigation outcomes.

9 Limitations

MALTLS-22 is not evaluated because the project does not have a verified source and license path. OpTC is not evaluated as a formal enterprise case because verified provenance events are absent. Process and threat-intelligence views are disabled in formal results when the underlying fields are unavailable. The current reference implementation should therefore be read as a real-data methodological validation, not as a complete enterprise deployment package.

10 Conclusion

Graph-CoLD extends CoLD-style noisy-label intrusion detection with multi-view graph context, a label-space consistency diagnostic, and evidence-preserving soft weights. In the verified CICIDS-2017 and CESNET-TLS-Year22 settings, Graph-CoLD improves paired Macro-F1 over CoLD and preserves more clean informative evidence than hard deletion. The next technical step is broader faithful baseline coverage and a verified enterprise provenance case; final journal upload additionally requires human author-metadata confirmation.

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Declaration of competing interest

No competing interest is recorded in this repository candidate package. The named authors must confirm the final declaration before journal upload.

Data and code availability

Raw datasets are not committed to the repository. The reproducibility package documents local data roots, audit gates, frozen result hashes, and commands for regenerating D5/D5.5, D6 tables and figures, and this manuscript.

Declaration of generative AI and AI-assisted technologies

A generative AI coding assistant was used during repository engineering to draft code, tests, audit reports, and manuscript-risk text. The authors must review and accept responsibility for all final manuscript content before journal upload.

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